

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 6: Stalling Mechanics, Boundary Layer & Flow Separation

Syllabus Source: CAE Oxford (pp. 185-224) | Pass Target: $\geq 90\%$ AviationExam

1. THE BOUNDARY LAYER: LAMINAR VS TURBULENT

The boundary layer is the microscopic fluid layer adjacent to the solid surface where viscous forces slow velocity from zero (surface no-slip) up to 99% of freestream:

Boundary Layer Characteristic	Laminar Boundary Layer	Turbulent Boundary Layer
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Fluid Particle Motion	Smooth, parallel microscopic streamlines with zero cross-mixing.	Vortical, swirling eddies with intense kinetic momentum exchange.
Layer Thickness & Profile	Extremely thin; steep velocity gradient at wall.	Thicker; flatter velocity profile with energetic fluid near wall.
Skin Friction Drag	VERY LOW (ideal for laminar flow airfoils).	HIGHER (roughly double the friction drag of laminar).
Separation Resistance	POOR. Lacks kinetic energy; separates readily under mild adverse pressure gradient.	EXCELLENT. Kinetic mixing energizes lower layers, delaying flow separation against adverse pressure gradients!

- **Transition Point:** The point where the boundary layer transitions from laminar to turbulent. Moves FORWARD with: increasing airspeed/Reynolds number, increasing angle of attack, adverse pressure gradient, and surface roughness.

2. AERODYNAMIC STALLING MECHANICS

As angle of attack increases past the maximum lift coefficient (C_{L_max}), the adverse pressure gradient on the upper surface decelerates boundary layer air to a complete standstill, reversing flow direction and causing MASSIVE FLOW SEPARATION:

- **Critical Angle of Attack (α_{crit}):** The stalling angle of attack (typically 15° to 18°). An airfoil ALWAYS stalls at the EXACT SAME critical angle of attack, regardless of airspeed, gross mass, bank angle, or altitude!
- **Stall Speed Formula:** $V_S = \sqrt{(2 \times W) / (\rho \times S \times C_{L_max})}$.
- **Factors Affecting Stall Speed:** 1. Mass: V_S increases proportional to $\sqrt{W}$; 2. Load Factor (n): $V_S = V_{S0} \times \sqrt{n} = V_{S0} \times \sqrt{1 / \cos(\text{bank})}$; 3. Center of Gravity: Forward CG increases stall speed (larger download on tailplane increases effective wing loading); 4. Flaps: Flap extension increases C_{L_max} , lowering stall speed.

[!] EXAM TRAP (AVIATIONEXAM) - Accelerated Stall & Steep Turns Calculation Trap

- In a level coordinated turn, Load Factor $n = 1 / \cos(\text{Bank Angle } \Phi)$:
 - At 45° Bank: $n = 1.41 \text{ g}$ -> Stall Speed increases by $\sqrt{1.41} = +19\%$!
 - At 60° Bank: $n = 2.00 \text{ g}$ -> Stall Speed increases by $\sqrt{2.00} = 1.414 (+41.4\%)$!
 - If an aircraft stalls at 60 kt clean: At 60° bank it will stall at $60 \times 1.414 = 85 \text{ KNOTS!}$

3. WING PLANFORM STALL PROGRESSION PATTERNS

- Rectangular Wing: Stalls at WING ROOT first. Safest planform: root stall buffet provides natural pilot warning; ailerons at wingtips remain in unstalled air, retaining full roll control.
- Tapered / Swept Wing: Stalls at WINGTIP first. Dangerous: tip stall causes loss of aileron roll control and creates a severe, uncommanded NOSE-UP pitching moment (washout twist and stall strips are added to force root stall).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: If an aircraft with a clean level-flight stall speed of 70 knots enters a coordinated 60° banked turn, what is the stall speed?

[A] 70 knots [B] 85 knots [C] 99 knots ($70 \times \sqrt{2}$) [D] 140 knots

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [C]. Load factor at 60° bank is $n = 1 / \cos(60^\circ) = 2.0$ g. The stalling speed formula is $V_{S(\text{turn})} = V_S \times \sqrt{n} = 70 \times \sqrt{2} = 70 \times 1.414 = 98.98$ (~99 knots).

Q2: Why does a TURBULENT boundary layer resist flow separation better than a LAMINAR boundary layer?

[A] Turbulent flow has zero viscosity [B] Turbulent vortices mix high-energy air from the outer freestream down into the lower layer, energizing it to overcome adverse pressure gradients [C] Turbulent flow is thinner [D] Turbulent flow has lower skin friction

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The defining advantage of turbulent boundary layers is kinetic turbulent mixing. High-speed fluid from above is churned down near the wall, allowing the flow to remain attached longer against opposing adverse pressure gradients.

Q3: How does moving the aircraft Center of Gravity (CG) FORWARD affect the clean stall speed?

[A] Reduces stall speed [B] INCREASES stall speed slightly [C] Has zero effect on stall speed [D] Causes an unrecoverable spin

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Forward CG increases the nose-down moment, requiring a greater downward aerodynamic force from the tailplane for trim. This extra downward load adds to the aircraft weight, increasing wing loading and raising stall speed.